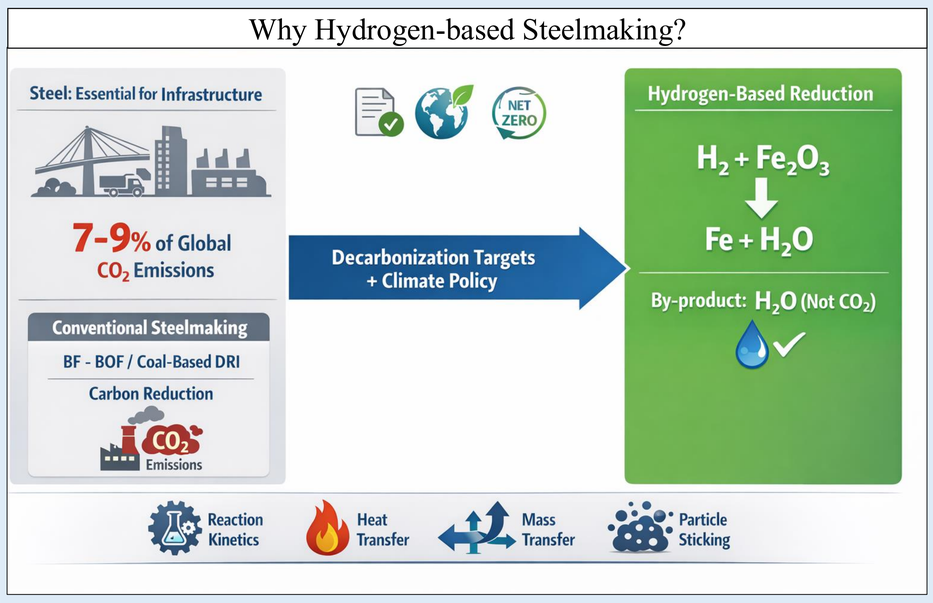
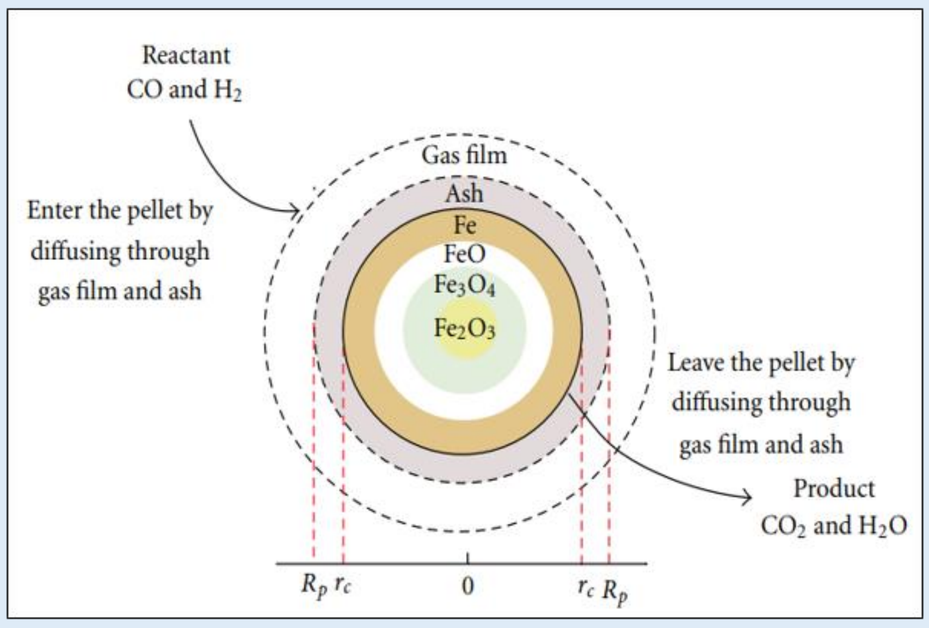
Iron ore reduction in an industrial rotary kiln using a sustainable pathway

Need of H2-based Iron ore reduction for making Green Steel



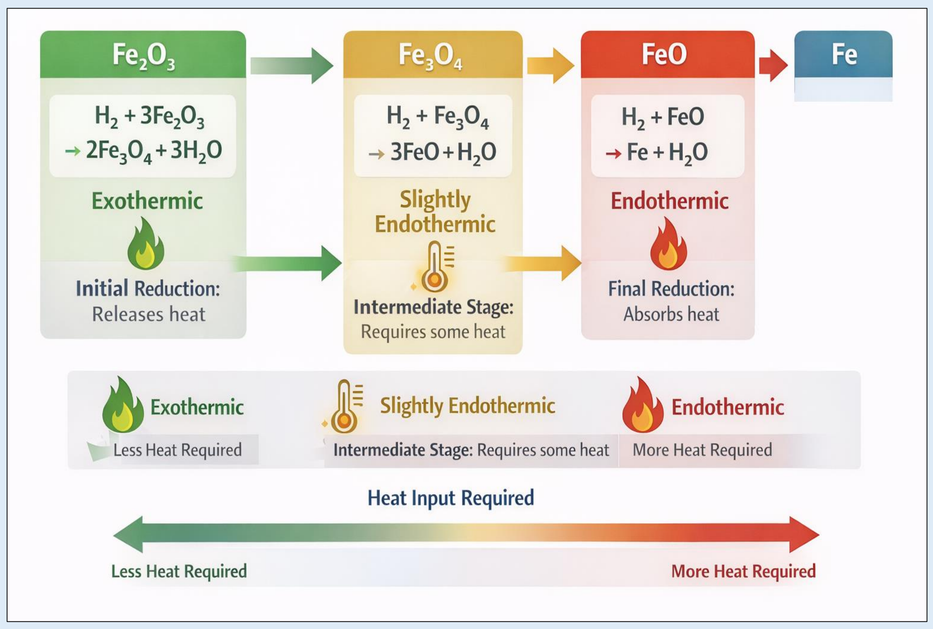
- Industrial-scale hydrogen reduction in rotary kilns is not fully understood.
- Gas–solid interaction and heat transfer strongly influence reduction efficiency.
- Need detailed insight into reaction–diffusion coupling.

Basics of iron ore reduction – Shrinking Core Model

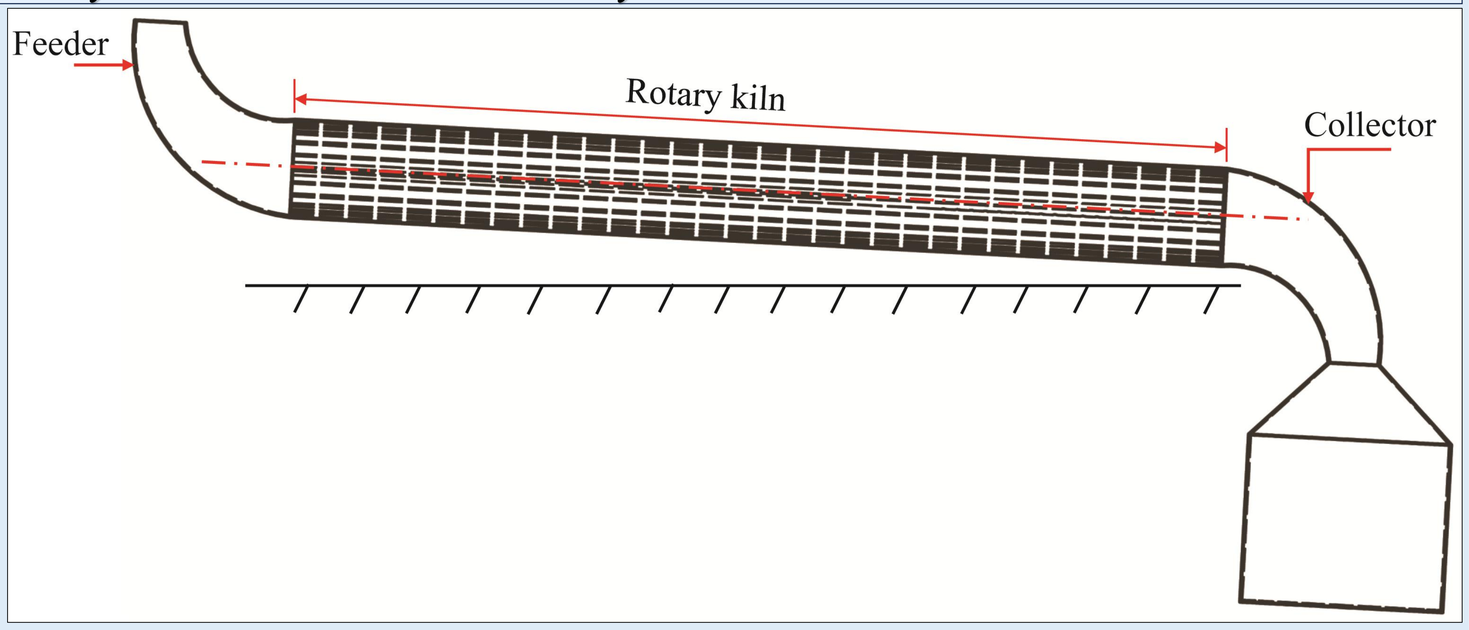


- **Layer-by-layer reduction:** Reaction progresses from surface to core, forming a porous iron shell while the unreacted Fe_2O_3 core shrinks with time.
- **Inward diffusion:** H_2 penetrates through the external gas film and porous product layer to reach the reaction interface.
- **Outward diffusion:** Generated H_2O diffuses outward through the porous iron layer into the bulk gas stream.

Reaction pathway and an Industrial Rotary kiln

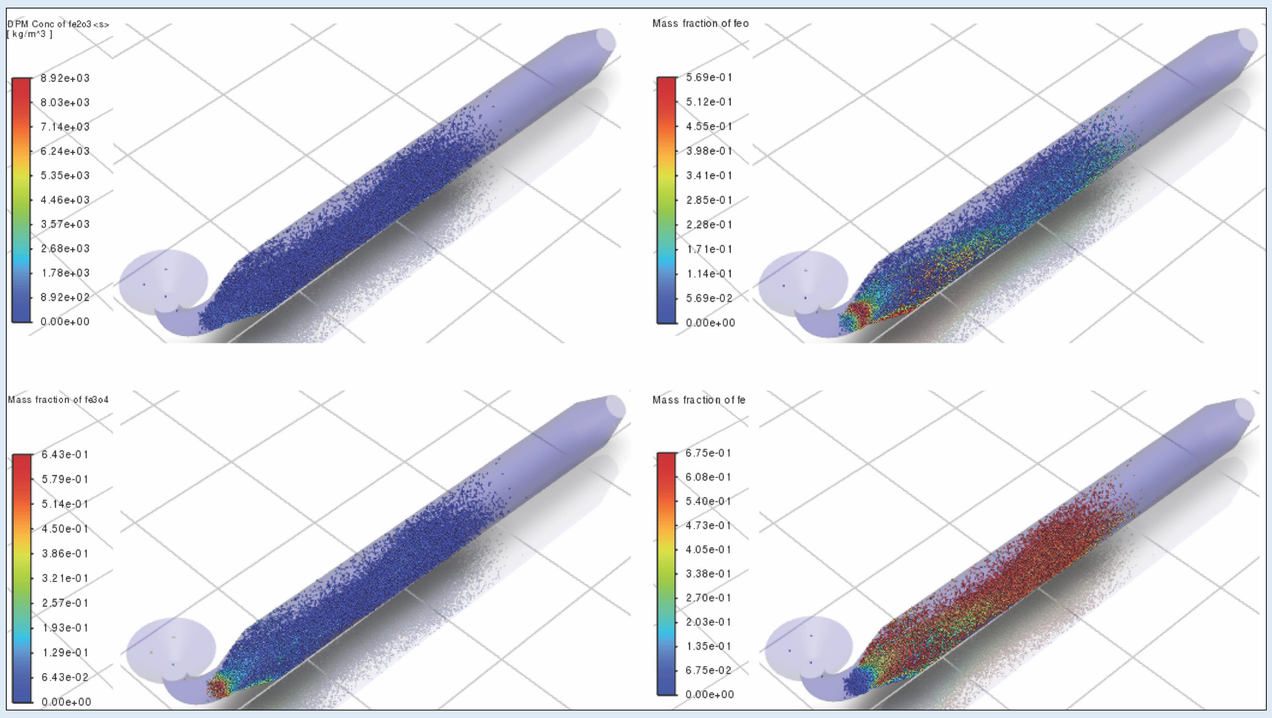


Reaction pathway for H2-based reduction



Schematic presentation of an Industrial Rotary kiln

CFD study to visualize DRI using H2 and Iron ore before and after reduction – Experiment



Before reduction

After reduction

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